

EXPERIMENT 2

MULTIPLICATION OF TWO 3×3 MATRICES

Objectives

- a) To write a Scilab program to perform the multiplication of the following two 3×3 matrices using nested loops:

$$\mathbf{A} = \begin{bmatrix} -9 & 2 & 5 \\ 3 & 3 & -4 \\ 7 & 5 & 1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & -6 & -2 \\ 5 & 5 & 8 \\ 2 & -3 & 4 \end{bmatrix}$$

- b) To verify the numerical result using Scilab's built-in matrix multiplication operator.

Theory

Matrix multiplication is a fundamental operation in linear algebra used extensively in physics for coordinate transformations and solving systems of linear equations.

For two matrices $\mathbf{A}$ and $\mathbf{B}$, the product matrix $\mathbf{C} = \mathbf{A} \times \mathbf{B}$ is defined if the number of columns in $\mathbf{A}$ is equal to the number of rows in $\mathbf{B}$. For 3×3 matrices, each element C_{ij} of the resultant matrix is calculated as the dot product of the i^{th} row of $\mathbf{A}$ and the j^{th} column of $\mathbf{B}$:

$$C_{ij} = \sum_{k=1}^3 A_{ik} B_{kj}$$

In Scilab, while the $*$ operator performs this operation directly, implementing it via loops demonstrates the underlying algorithmic logic.

Scilab Code

```
1 function C = matrix_mult(A, B)
2     [m, n] = size(A);
3     [p, n] = size(B);
4     if n ~= p then
5         error("Inconsistent dimensions for multiplication");
6     else
7         C = zeros(m, n); // Initialize resultant matrix with zeros
8         for i = 1:m
9             for j = 1:n
10                 for k = 1:p
11                     C(i,j) = C(i,j) + A(i,k) * B(k,j);
12                 end
13             end
14         end
15     end
16 endfunction
17
18 A = [-9, 2, 5; 3, 3, -4; 7, 5, 1];
19 B = [1, -6, -2; 5, 5, 8; 2, -3, 4];
20
21 // Perform multiplication using the function
22 C_manual = matrix_mult(A, B);
23
24 // Perform multiplication using built-in operator
25 C_builtin = A * B;
26
27 // Check differences in results
28 C_diff = C_manual - C_builtin;
29
30 // Display Results
31 disp("Matrix A:", A);
32 disp("Matrix B:", B);
33 disp("Product (Manual Loop):", C_manual);
34 disp("Product (Built-in):", C_builtin);
35 disp("Difference matrix:", C_diff);
```

Output

Console output of Scilab code:

Scilab 2025.1.0 Console

```
--> exec('D:\Professional\Work\RC\courses\Lab\Sem-04_Math-Phys\expt-02.sce', -1)
"Matrix A:"
-9.  2.  5.
 3.  3. -4.
 7.  5.  1.
"Matrix B:"
 1. -6. -2.
 5.  5.  8.
 2. -3.  4.
"Product (Manual Loop):"
11. 49. 54.
10.  9.  2.
34. -20. 30.
"Product (Built-in):"
11. 49. 54.
10.  9.  2.
34. -20. 30.
"Difference matrix:"
 0.  0.  0.
 0.  0.  0.
 0.  0.  0.
--> |
```

Variable Browser

Name	Value	Type	Visibility	Memory
A		3x3 Double	local	280 B
B		3x3 Double	local	280 B
C_builtin		3x3 Double	local	280 B
C_diff		3x3 Double	local	280 B
C_manual		3x3 Double	local	280 B

Command History

```
scf(1); plot(x, y_sol1(1, :));
scf(2); plot(x, y_sol2(1, :));
scf(3); plot(x, y_sol3(1, :));
scf(4); plot(x, y_sol4(1, :));
scf(1); plot(x, y_sol1(1, :));
title('$\text{Plot of } \int_0^x \sin(t) dt$')
title('$\text{Solution of } \frac{dy}{dx} = e^x$')
xlabel('$x$')
xlabel('$x$', 'fontsize', 12)
xlabel('$x$', 'fontsize', 12)
xlabel('$x$')
xlabel('$x$', 'fontsize', 12)
xlabel('$x$', 'fontsize', 6)
xlabel('$x$', 'fontsize', 4)
xlabel('$x$', 'fontsize', 5)
```

Results

- The multiplication of the given matrices was performed using nested loops in Scilab.
- The numerical result was verified by comparing with the output from Scilab's built-in matrix multiplication operator.